

Hardware Design and Component Specification for an ESP32-Based Multi-Channel 2.4 GHz and Sub-GHz RF Spectrum Analyzer and Jammer Platform

Architectural Rationale and System Concept

Modern wireless security auditing and spectral monitoring require highly integrated, portable hardware architectures capable of analyzing and interacting with diverse radio frequency (RF) protocols¹. Developing a multi-protocol diagnostic platform involves utilizing an Espressif ESP32 microcontroller as the central computational unit, complemented by an array of specialized RF transceivers². The primary objective of such a design is to bridge the gap between high-frequency digital signal processing and localized RF physical-layer manipulation¹. To achieve multi-band coverage across the 2.4 GHz Industrial, Scientific, and Medical (ISM) band and various sub-GHz frequencies, the system architecture employs a modular transceiver array¹. A single transceiver module is physically limited by its frequency synthesizer, which can only lock onto and monitor one frequency channel at any given instant⁷. To overcome this hardware limitation and enable simultaneous multi-channel sniffing, signal analysis, and active defensive signal generation, the platform integrates multiple Nordic Semiconductor nRF24L01+ modules alongside a Texas Instruments CC1101 sub-GHz transceiver¹.

This multi-module configuration allows the system to execute parallel tasks, such as dedicated spectrum sweeping on one module while simultaneously performing targeted packet sniffing or signal disruption on others¹. The ESP32 is highly suited for coordinating this array due to its dual-core Xtensa processor architecture, which allows developers to decouple timing-critical SPI RF driver routines on one core from user-interface rendering, file system access, and sensor logging on the other⁹.

Technical Specifications and Component Allocation

Executing a professional-grade hardware design requires a detailed, verified Bill of Materials (BOM)¹¹. The selected components must balance electrical compatibility, thermal limits, and physical footprint requirements⁷. The following table compiles the exact hardware components, passive filtering elements, and expansion sensors necessary to construct a fully integrated spectrum analyzer and testing platform⁹.

Component Category	Description / Manufacturer Part	Quantity	Operational Role	Key Technical Specifications and Selection Criteria
Microcontroller	Espressif ESP32-WROO M-32U ¹¹	1	Central system processing, GUI rendering, and native Wi-Fi/BLE auditing ⁹ .	Dual-core 240 MHz processor ² ; features an integrated U.FL connector for external high-gain antennas ¹¹ .
RF Transceiver (2.4 GHz)	Nordic nRF24L01+PA+LNA ⁵	3 or 4 ¹	2.4 GHz spectrum scanning, packet sniffing, and signal disruption ¹ .	Features a logic-controlled power amplifier and low-noise amplifier; +20 dBm peak output ⁵ .
RF Transceiver (Sub-GHz)	Texas Instruments CC1101 Breakout ⁴	1	Sub-GHz protocol analysis, signal replay, and IoT monitoring ² .	Operates across 315, 433, 868, and 915 MHz; support for configurable FSK/GFSK/OOK modulation ⁷ .
Display Module	ILI9341 2.4-inch TFT LCD with SPI Touch ⁵	1	Real-time spectrum visualization and user menu navigation ⁵ .	SPI-controlled 240 × active matrix; includes an on-board MicroSD slot sharing SPI

				lines ⁹ .
Power Converter	MP1584EN Synchronous Step-Down ¹²	1	High-efficiency primary step-down from battery/USB to 5.0V ¹² .	Supports input voltages up to 28V ; delivers up to 3A with efficiency exceeding 90% ¹² .
Linear Regulator	MIC5219-3.3 Low-Dropout Linear Regulator ¹¹	2	Clean, low-noise 3.3V rail dedicated to the sensitive RF modules ¹² .	Ultra-low-noise output with 500 mA capability; isolates RF transceivers from digital logic noise ¹² .
Battery Management	TP4056 Linear Li-Ion Charger with Protection ⁹	1	Safe linear charging and over-discharge protection for 18650 cells ⁹ .	Features dual LED status indicators and built-in protection against cell over-discharge below 2.4V ¹¹ .
Storage Module	SPI-based MicroSD Card Slot ⁴	1	Local logging of captured signal packets, PCAP files, and configuration data ⁴ .	Interface wired to share hardware SPI with the display, requiring strict tri-state buffer control ¹⁸ .
Geospatial Sensor	Neo-6M GPS Module ⁹	1	Real-time coordinate capture for	Communicates via hardware UART; tracks

			spatial mapping of spectrum data ⁹ .	up to 50 satellite channels with a 1 Hz update rate ⁹ .
Environmental Sensor	BME280 Temperature, Humidity & Pressure ¹⁵	1	Atmospheric data logging during long-term field auditing ¹¹ .	Communicates over I2C; low current draw of 3.6μA at 1 Hz sampling frequency ¹¹ .
Voltage Sensor	MAX17043 LiPo Fuel Gauge	1	High-precision battery voltage and state-of-charge tracking	Communicates over I2C; uses a ModelGauge algorithm to eliminate the need for sense resistors.
Optical Sensor	TEMT6000 Ambient Light Sensor	1	Automatic screen backlight brightness adjustment to optimize power	Analog output mapped to an ESP32 ADC pin; peak sensitivity matches the human eye.
Visual Indicator	WS2812B NeoPixel RGB LED ¹	1	Status and mode signaling via configurable color outputs ¹	Single-wire digital control; cascades easily and draws minimal power during idle states ¹¹ .
User Inputs	Tactile Pushbuttons ¹¹	5	Physical menu navigation (Up,	Interfaced to GPIOs

			Down, Left, Right, Select) ¹¹	configured with internal pull-up resistors and software debounce ¹¹ .
Passive Filters	Ceramic Decoupling Capacitors ¹²	Multiple	High-frequency noise suppression close to transceiver VCC pins ¹² .	0.1μF and 10μF multi-layer ceramic capacitors (MLCC) with low equivalent series resistance ⁵ .
Passive Filters	Bulk Electrolytic Capacitors ¹²	4	Buffering power draw during high-gain transmission transients ¹² .	220μF low-ESR radial electrolytic capacitors rated to at least 6.3V ²⁴ .

Circuit Schematic and Pin Mapping

To ensure proper communication and prevent pin multiplexing conflicts, the physical connections must be planned in accordance with the ESP32's internal hardware routing capabilities¹⁰. The display and storage peripherals are mapped to the VSPI hardware bus, while the RF transceivers utilize the HSPI bus²⁶. The following table provides the precise pin mappings for the entire system, establishing an unambiguous wiring schematic⁵.

Peripheral Module	Physical Pin on Module	ESP32 GPIO Destination	Signal Designation	Hardware Interface Details and Configuration Notes

nRF24L01+ Module #1 [cite: 27]	VCC ¹⁶	Dedicated LDO Output	3.3V _{RF}	Powered from the clean, isolated analog regulator rail ¹² .
	GND ¹⁶	Common Ground	GND	Connected directly to the system-wide ground plane ¹² .
	SCK ¹⁶	GPIO 14 ²⁷	HSPI_SCLK ²⁶	Dedicated hardware clock line for RF operations ²⁶ .
	MOSI ¹⁶	GPIO 23 ²⁷	HSPI_MOSI ²⁶	Master Out Slave In line routing data to transceivers ²⁶ .
	MISO ¹⁶	GPIO 19 ²⁷	HSPI_MISO ²⁶	Master In Slave Out line capturing data from transceivers ²⁶ .
	CE ¹⁶	GPIO 5 ²³	RF1_CE ²³	Configured as a digital output to enable module RX/TX ⁸ .
	CSN ¹⁶	GPIO 17 ²³	RF1_CSN ²³	Active-low Chip Select; kept HIGH during idle states ⁸ .
nRF24L01+ Module #2 [cite: 27]	CE	GPIO 16 ²⁷	RF2_CE	Configured as a digital output to enable module

				RX/TX ⁸ .
	CSN	GPIO 4 ²⁷	RF2_CSN	Active-low Chip Select; kept HIGH during idle states ⁸ .
nRF24L01+ Module #3 [cite: 27]	CE	GPIO 15 ²⁷	RF3_CE	Configured as a digital output to enable module RX/TX ⁸ .
	CSN	GPIO 2 ²⁷	RF3_CSN	Active-low Chip Select; kept HIGH during idle states ⁸ .
CC1101 Sub-GHz Transceiver [cite: 4]	CSN ¹⁷	GPIO 21	CC_CSN	Active-low Chip Select isolating Sub-GHz register writes ¹⁷ .
	GDO0 ¹⁷	GPIO 22	CC_GDO0	Interrupt pin indicating payload receipt or TX complete ¹⁷ .
ILI9341 TFT Display [cite: 5, 9]	SCLK	GPIO 18	VSPI_SCLK ²⁶	Shared high-speed clock for display rendering ²⁶ .
	MOSI	GPIO 23 ⁵	VSPI_MOSI ²⁶	Shared data line routing frame data to

				the screen ²⁶ .
	MISO	GPIO 19 ⁵	VSPI_MISO ²⁶	Shared data return line; active during register reads ²⁶ .
	CS ⁵	GPIO 15 ⁵	TFT_CS	Active-low line enabling SPI communication with display ⁵ .
	D/C	GPIO 21	TFT_DC	Data/Command selection line for display register control.
	RST	GPIO 22	TFT_RST	Hardware reset pin to initialize display controller on boot.
MicroSD Card Module [cite: 4, 11]	CS	GPIO 13	SD_CS	Active-low Chip Select; requires MISO isolation ¹⁸ .
Neo-6M GPS Module [cite: 9]	TX	GPIO 34	GPS_RX	Hardware UART receiver channel mapped to ESP32 ADC/Input.
	RX	GPIO 35	GPS_TX	Hardware UART transmitter channel.

Environmental Sensor (BME280)	SDA	GPIO 21	I2C_SDA ²⁷	Hardware I2C data line; requires 4.7 kΩ pull-up resistors ¹¹ .
	SCL	GPIO 22	I2C_SCL ²⁷	Hardware I2C clock line; requires 4.7 kΩ pull-up resistors ¹¹ .
Tactile Buttons (5-way array)	UP ²⁷	GPIO 26 ²⁷	BTN_UP ²⁷	Pin configured as input with internal pull-up resistor ¹¹ .
	SELECT ²⁷	GPIO 32 ²⁷	BTN_SEL	Pin configured as input with internal pull-up resistor ¹¹ .
	DOWN ²⁷	GPIO 33 ²⁷	BTN_DOWN ²⁷	Pin configured as input with internal pull-up resistor ¹¹ .
	LEFT	GPIO 25 ²⁷	BTN_LEFT	Pin configured as input with internal pull-up resistor ¹¹ .
	RIGHT	GPIO 27 ²⁷	BTN_RIGHT	Pin configured as input with internal pull-up resistor ¹¹ .
NeoPixel WS2812B	DIN	GPIO 12	WS2812_DATA	Single-wire high-speed

[cite: 1, 11]				timing-sensitive data interface ¹¹ .
Ambient Light Sensor	OUT	GPIO 36	ALS_ANALOG	Analog output read via internal 12-bit ADC Channel 1.

RF Signal Analysis and Jamming Mechanisms

The primary operation of the diagnostic tool is split between passive signal analysis and active transmission testing, which can function as a jammer for vulnerability research¹. The underlying physics of these operations dictate distinct software execution models and strict transceiver state transitions¹⁷.

Spectrum Analysis Mechanics

To construct a real-time visualization of the 2.4 GHz spectrum, the platform leverages the nRF24L01+ Received Power Detector (RPD) register²⁹. Register 0x09 contains a single-bit field at index 0 that reflects whether the power level in the currently selected channel exceeds the hardware threshold of -64 dBm ¹⁶. The scanning sweep routine is executed sequentially across 128 channels, spaced at 1 MHz increments starting from 2.400 GHz up to 2.525 GHz ⁸.

For each channel step, the firmware writes the channel offset to the RF_CH register (0x05), toggles the Chip Enable (CE) pin high for approximately $130 \mu\text{s}$ to allow the phase-locked loop (PLL) frequency synthesizer to lock and stabilize, and then reads the RPD register⁸. If the RPD bit is returned as high, a signal level exceeding -64 dBm is present, and the hit counter for that specific channel increment is updated in the display frame buffer³⁰.

Integrating multiple nRF24L01+ modules allows the sweep workload to be divided¹. Assigning Module 1 to sweep channels 0–42, Module 2 to sweep channels 43–85, and Module 3 to sweep channels 86–127 reduces the total scan period by a factor of three, resulting in high-speed, real-time waterfall updates on the UI display¹.

Active Jamming Methodologies

When configured in active transmission mode, the system acts as a signal jammer to test the robustness of wireless protocols¹. Physical layer jamming can be achieved through two distinct approaches¹:

- **Constant Carrier Wave Injection:** The transceiver is programmed into a non-standard

test state where the PLL frequency synthesizer is locked onto a target channel while packet framing, preamble generation, and cyclic redundancy checks (CRC) are bypassed¹⁷. This forces the module to transmit a continuous, unmodulated RF carrier

wave at maximum power (**+20 dBm**) on the target channel, raising the noise floor and preventing nearby receivers from decoding valid packets¹⁶.

- **Pseudo-Random Frequency Hopping Disruption:** Modern protocols like Bluetooth and Wi-Fi utilize Frequency Hopping Spread Spectrum (FHSS) or Orthogonal Frequency Division Multiplexing (OFDM) to avoid static channel noise⁷. To disrupt these agile signals, the ESP32 orchestrates the multi-transceiver array to sweep rapidly across a wide frequency band¹. Module 1 can be configured to continuously flood BLE advertising channels (37, 38, and 39) with random, invalid packet headers¹. Simultaneously, Modules 2 and 3 can be commanded to sweep pseudo-randomly across Wi-Fi channel bands, injecting high-rate noise payloads to trigger packet collisions and force continuous retransmission attempts by target devices².

Electromagnetic Compatibility, Power Filtering, and SPI Bus Coexistence

Combining high-frequency digital microcontrollers, multi-layer liquid crystal displays, high-power RF transmitters, and localized flash memory onto a single development platform introduces significant signal integrity issues⁹. Managing these issues requires addressing power supply noise and SPI bus contention¹⁸.

Power Supply Noise and Thermal Dissipation

The current consumption profile of the platform is highly dynamic⁹. While the ESP32 averages approximately **100 mA** during processing, it can spike to **250 mA** when transmitting native Wi-Fi beacons³. Simultaneously, each nRF24L01+PA+LNA transceiver draws up to **115 mA** during peak power packet transmissions¹⁶.

Under worst-case scenarios where three nRF24 modules and the sub-GHz CC1101 transmit simultaneously alongside the active TFT display backlight and high-speed SD card writes, the aggregate peak load is calculated as⁷:

$$I_{\text{peak}} = I_{\text{ESP32}} + 3(I_{\text{nRF_TX}}) + I_{\text{CC1101}} + I_{\text{Display}} + I_{\text{SD_Write}}$$

$$I_{\text{peak}} = 250 \text{ mA} + 3(115 \text{ mA}) + 35 \text{ mA} + 100 \text{ mA} + 100 \text{ mA} = 830 \text{ mA}$$

To step down an input voltage of **12V** (typical of standard external wall adapters) to the required **3.3V** system rail, a simple linear regulator would be highly inefficient¹². The thermal

power dissipation (P_{diss}) of a linear regulator is expressed as³⁷:

$$P_{\text{diss}} = (V_{\text{in}} - V_{\text{out}}) \times I_{\text{out}}$$

$$P_{\text{diss}} = (12\text{V} - 3.3\text{V}) \times 0.83\text{A} = 7.221\text{W}$$

Given that a standard SOT-223 regulator has a junction-to-ambient thermal resistance (θ_{JA}) of approximately 136°C/W without extensive heat sinks, a power dissipation of over 7W would instantly trigger thermal shutdown or cause hardware failure³⁵. To resolve this, the design utilizes a dual-stage regulation topology¹². A synchronous buck converter (MP1584EN) is placed at the primary input stage to step down the external voltage (5V to 12V) to an intermediate 5.0V rail with up to 95% conversion efficiency¹². This intermediate 5.0V rail then feeds two independent, low-noise LDO linear regulators¹¹. The first LDO regulates 3.3V for the digital logic, display, and SD card¹². The second LDO is dedicated entirely to the sensitive analog RF transceivers¹². This isolates high-frequency digital noise and preventing voltage drops during SD card writes from corrupting the RF ground plane¹⁷.

SPI Bus Contention and Level Shifter Lockup

Integrating multiple high-speed SPI devices onto the ESP32's hardware buses can lead to bus lockups¹⁰. A common point of failure involves the level-shifter integrated circuits (such as the 74HC125) typically included on cheap MicroSD card breakout modules¹⁸. These breakout modules are designed to convert 5V microcontroller signals to the 3.3V required by the SD card¹⁸. However, many of these boards feature an architectural flaw: the active-low Output Enable ($\overline{\text{OE}}$) pin of the level-shifter chip is tied directly to the ground plane¹⁸. This causes the level-shifter's output buffers to drive the shared Master In Slave Out (MISO) line continuously, even when the SD card's Chip Select (CS) line is pulled HIGH to deselect the module¹⁸. This prevents other peripherals on the shared SPI bus (such as the nRF24 transceivers) from tri-stating their MISO outputs and communicating with the ESP32¹⁸. To prevent this MISO lockup, two hardware solutions can be implemented¹⁸:

- **Physical Circuit Modification:** Modifying the MicroSD card breakout module involves scraping the PCB trace connecting the level-shifter's pin 1 and pin 4 ($\overline{\text{OE}}$ buffers) to ground, and soldering a jumper wire directly from those pins to the SD card Chip Select line²¹. This ensures that when the SD CS line is pulled high to deselect the card, the level-shifter buffers enter a high-impedance tri-state, freeing the shared MISO line¹⁸.

- **Dedicated Software SPI Isolation:** Rather than placing the SD card and the display on the same hardware SPI bus as the sensitive transceivers, the storage card can be isolated onto separate GPIO pins using the SdFat library's software SPI (SoftSPI) implementation¹⁸. Since diagnostic logging is typically less latency-sensitive than RF packet capture, dedicating the ESP32's primary hardware HSPI and VSPI buses to the display and the transceiver array optimizes signal timing¹⁰.

Regulatory Compliance and Legal Frameworks

The operation of RF-emitting apparatus, particularly devices configured for signal analysis and active transmission testing, is heavily governed by national and international telecommunications laws³⁹. Operating unlicensed transmitters or emitting active interference can result in severe legal consequences³⁹.

Telecommunications Regulatory Commission of Sri Lanka (TRCSL) Framework

In Sri Lanka, the Telecommunications Regulatory Commission (TRCSL) is the statutory body responsible for spectrum management and licensing under the Sri Lanka Telecommunications Act No. 25 of 1991⁴⁰.

The TRCSL enforces several strict regulatory compliance mandates:

- **Importation and Possession Bans:** Under Section 36 of the consolidated telecommunications statutes, it is a criminal offence to import, possess, or operate any wireless telegraphy apparatus or terminal equipment without an authorized licence of the prescribed description issued directly by the regulatory authority⁴¹.
- **Type Approval Requirements:** Under Section 5 of the Act, all Radio and Telecommunications Terminal Equipment (RTTE) must undergo a formal Type Approval process before importation, sale, or commercial deployment⁴⁰. Standard Type Approval requires submitting laboratory test reports from accredited institutions to verify that the equipment complies with electromagnetic compatibility (EMC), radio frequency (RF) exposure, and health standards⁴⁵.
- **Low-Power Device Exemptions:** The TRCSL provides exemptions from individual station licensing for certain low-power, short-range devices (SRDs)⁴⁰. However, these devices must operate strictly within designated bands and must not exceed specified Equivalent Isotropically Radiated Power (EIRP) limits, typically set at **23 dBm (200 mW)** for LPWAN bands or under **50 mW** for standard short-range Bluetooth and Wi-Fi devices⁴⁰.

Enforcement, Penalties, and Spectrum Surveillance

The TRCSL operates a network of fixed and mobile spectrum monitoring and direction-finding (DF) stations across Sri Lanka, capable of monitoring frequencies from **10 kHz** to **24 GHz**³⁹. These stations are designed to identify, isolate, and locate unauthorized radio emissions,

illegal transmitters, and active signal jammers³⁹. Under Section 39 of the Sri Lanka Telecommunications (Amendment) Act No. 39 of 2024 and consolidated regulatory statutes, the legal penalties for unlicensed operation and spectrum interference are severe⁴³:

- **Fines and Imprisonment:** Any individual or operator who installs, possesses, or works an unlicensed radio transmitter or causes active, harmful interference to licensed commercial, emergency, or military communications networks commits an offence³⁹. Upon conviction by a Magistrate, the offender is liable to a fine not exceeding ten million rupees, imprisonment for a term not exceeding one year, or both⁴⁷.
- **Asset Seizure:** Regulatory authorities, supported by law enforcement and the Ministry of Defence, have the power to enter properties, seize illegal transmitters, and permanently confiscate all associated computer systems, power equipment, and antenna structures⁴¹.
- **Active Jamming Prohibition:** Active signal jamming devices (which block cellular networks, GPS, Wi-Fi, or emergency channels) are strictly illegal under civil frameworks worldwide¹. Because these devices can disrupt critical infrastructure and emergency services, their deployment is classified as a severe threat to public safety and national security³⁹.

Therefore, any experimental verification of the RF transmission, packet injection, or signal disruption mechanisms described in this design must be conducted strictly within a shielded laboratory environment¹. This includes the use of sealed Faraday cages or calibrated RF anechoic chambers to ensure that no signals escape into the public spectrum, thereby maintaining compliance with national and international laws³⁹.

Architectural Synthesis and Implementation Summary

To assist developers in avoiding common hardware integration issues, the final system assembly should align with the structural recommendations consolidated in the following table².

Design Challenge	Primary Cause	Mitigation Strategy and Architectural Implementation
RF Transmit Failure [cite: 24, 25]	Voltage drops under transient load peaks during transmission ¹² .	Solder a low-ESR 10μF ceramic and 220μF electrolytic capacitor directly to the VCC/GND pins of each transceiver ⁵ .
Self-Interference / EMI	High-frequency noise from	Utilize metal shield cans on

[cite: 25, 52]	unshielded PA/LNA transceivers ²⁵ .	RF modules and route the ESP32's on-board antenna away from transceiver antennas ⁷ .
SPI Bus Freeze [cite: 18, 21]	Level-shifter output conflict on cheap SD card breakout boards ¹⁸ .	Modify the card slot PCB to connect level-shifter enable pins directly to Chip Select, or isolate storage using Software SPI ¹⁸ .
Regulator Overheating [cite: 12, 37]	Excessive power dissipation dropping 12V to 3.3V via linear regulation ¹² .	Implement a dual-stage system using a switching buck converter for primary step-down, followed by low-noise linear regulators ¹² .
Regulatory Liability [cite: 39, 41]	Unauthorized signal emissions violating local laws ³⁹ .	Disable all active signal generation software modules during open-air testing; conduct all active transmission trials in a Faraday cage ¹ .

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